Role-Unit: A Discriminative Unit-Testing Benchmark for Functional Roles in Multi-Agent Systems

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1 Problem Statement

Multi-Agent Systems (MAS) are becoming the standard in AI development. In a MAS, multiple LLMs collaborate, with each taking on a different functional role. The industry is releasing new models constantly: some big, some small, some open-source, some closed. These models have vastly different capabilities. One model might be an expert at coding, while another is better at planning.

The main issue is matching the right model to the right role. Some roles are simple, like formatting output into JSON. Using a massive, expensive model for this is a waste of money. On the other hand, a model might be strong in general reasoning but lack the specific domain knowledge that a smaller, fine-tuned model has. Determining exactly which model qualifies for a specific role is critical for building efficient systems.

Existing solutions like RouteLLM [4] try to address this by routing queries to different models. However, they typically select from a fixed pool of LLMs and rely on RL training to optimize the selection. This is expensive to train and rigid; you cannot easily swap new models into the pool without re-training.

Benchmarking is also a bottleneck. Most agentic benchmarks, such as AgentBench [3], evaluate the entire workflow of a system, which hides the performance of individual roles. Standard LLM benchmarks for Math [1] or Reasoning [2] are mostly generative and simulation-based. They are slow to run, expensive, and non-deterministic. We currently lack a fast, deterministic "unit test" to verify if a model is fit for a specific role before deploying it.

2 Proposed Technique

To make the LLM roles unit testable, we propose a role based benchmarking system. The goal is to evaluate whether an LLM have the knowledge and skills to perform well in a specific role.

The one problem in making unit tests is the non-deterministic generative nature of LLMs. Also for a same question there can be multiple correct answers. To make our unit tests deterministic, we shift the approach from generative to discriminative. Instead of asking the LLM to write an answer, we will post a question and provide a series of options. The LLM would have to verify these answers and select the right one. By including distractors (wrong or very similar answers), we can check the model's role alignment strictly. The LLM have to choose which answer is the most correct.

To make it widely adaptable, we propose a python unit testing framework Role-Unit. The workflow is designed for developers:

- **Easy Usage:** Developers can download our package with `pip install role-unit`.
- **Standardized Testing:** The tool comes with our benchmark suite, allowing users to test any LLM from our set of roles.
- **Extensible:** Developers can add new unit tests to existing roles or add new roles using our framework.

To build the unit tests we will use existing datasets. For example, for maths where deterministic answer is possible, we use standard sets. For domain specific roles like a Legal Helper, we will test with exam questions from legal books or interview questions. That will be our strategy to collect and build the dataset.

Initially, we plan to add 10 roles with 30 handcrafted tests for each role to establish a baseline.

3 Expected Outcomes

We anticipate three primary contributions from this work, aiming to bridge the gap between model capabilities and role-specific requirements in Multi-Agent Systems (MAS).

(1) **The Benchmark Release:** We expect to release Role-Unit, a standardized benchmark designed specifically for unit-testing the functional roles of an LLM. Unlike general benchmarks, this will target the specific roles(e.g., Planner, Coder, Router) required in agentic workflows.
(2) **The Open-Source Framework:** To ensure usability, we will release a Python-based open-source framework. This tool will allow developers to easily import our testing suite and validate their own models against our role definitions with a simple API.
(3) **Correlation with Downstream Performance:** We hypothesize a strong correlation between our unit tests and real-world performance. We expect that models which achieve high accuracy on our discriminative Test will perform significantly better and more reliably in complex, generative MAS workflows.

Topic Number for Project

Group 1: Comprehensive Benchmarking of Large Language Models

References

[1] Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, Mark Chen, Heewoo Jun, Lukasz Kaiser, Matthias Plappert, Jerry Tworek, Jacob Hilton, Reiichiro Nakano, et al. 2021. Training Verifiers to Solve Math Word Problems. *arXiv preprint arXiv:2110.14168* (2021).
[2] Dan Hendrycks, Collin Burns, Steven Basart, Andy Zou, Mantas Mazeika, Dawn Song, and Jacob Steinhardt. 2021. Measuring Massive Multitask Language Understanding. In *International Conference on Learning Representations*.
[3] Xiao Liu, Hao Yu, Hanchen Zhang, Yifan Xu, Xuanyu Lei, Hanyu Lai, Yu Gu, Hangliang Ding, Kaiwen Men, Kejuan Yang, et al. 2024. AgentBench: Evaluating LLMs as Agents. In *The Twelfth International Conference on Learning Representations*.
[4] Isaac Ong, Simchon Amouyal, Chuhui Wu, Amit Mandelbaum, Christopher Potts, Matei Zaharia, and Omar Khattab. 2024. RouteLLM: Learning to Route LLMs with Preference Data. *arXiv preprint arXiv:2406.18665* (2024).